Mathematical Assignment 2:
M. Tech (CS), 2025 - 27

Instructions

Provide complete mathematical derivations. Justify every step rigorously.

Problem 1: Geometry of Quadratic Optimization

Consider the quadratic function

$$f(x) = \frac{1}{2}x^T Qx - b^T x,$$

where

$$Q = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad x \in \mathbb{R}^2.$$

(a) Show that Q is symmetric positive definite.

(b) Compute the unique global minimizer x^* analytically.

(c) Describe geometrically the level sets of $f(x)$.

(d) Compute eigenvalues and eigenvectors of Q and explain how they determine the shape of level curves.

Problem 2: Gradient Descent Dynamics

Consider gradient descent with constant step size η:

$$x_{k+1} = x_k - \eta \nabla f(x_k).$$

(a) Derive the explicit iteration formula in matrix form.

(b) Show that the error evolves as:

$$e_{k+1} = (I - \eta Q)e_k, \quad e_k = x_k - x^*.$$

(c) Derive the spectral radius condition for convergence.

(d) Determine the range of η for which gradient descent converges.

(e) Compute the optimal step size η_{opt} minimizing the asymptotic convergence factor.

Problem 3: Rate of Convergence and Condition Number

Let Q have eigenvalues λ_{min} and λ_{max}.

(a) Show that convergence rate depends on the condition number:

$$\kappa = \frac{\lambda_{max}}{\lambda_{min}}.$$

(b) Prove that the optimal convergence factor equals:

$$\rho^* = \frac{\kappa - 1}{\kappa + 1}.$$

(c) Interpret geometrically why ill-conditioning slows gradient descent.

(d) Suppose we perform a linear change of coordinates $y = P^T x$ where P diagonalizes Q. Show that in the new coordinates gradient descent decouples into scalar recursions.

Problem 4: Innovation — Designing a Worst-Case Quadratic

Construct a 2×2 positive definite matrix Q with condition number $\kappa = 100$.

(a) Choose eigenvalues and construct such a matrix.

(b) Analyze the gradient descent trajectory starting from $x_0 = (1, 1)^T$.

(c) Illustrate analytically why iterates zig-zag.

(d) Propose a preconditioning matrix M that improves convergence and justify mathematically.

Problem 5: Constrained Optimization with Two Equality Constraints

Minimize

$$f(x, y, z) = x^2 + y^2 + z^2$$

subject to

$$x + y + z = 1,$$

and

$$x - y + 2z = 0.$$

(a) Formulate the Lagrangian with two multipliers.

(b) Derive the full KKT system.

(c) Solve the system analytically.

(d) Verify that the solution is the global minimizer.

(e) Interpret the solution geometrically as projection onto an affine subspace.

Problem 6: Theoretical Extension

Let $f(x) = \frac{1}{2}x^T Q x - b^T x$ with Q symmetric positive definite.

(a) Prove that f is strongly convex.

(b) Show that gradient descent with $0 < \eta < \frac{2}{\lambda_{max}}$ converges linearly.

(c) Derive an explicit bound:

$$\|x_k - x^*\| \leq \rho^k \|x_0 - x^*\|.$$

(d) Compare gradient descent with Newton's method for this quadratic.

Problem 7: Decision Tree – Analytical and Numerical

Consider the dataset:

ID	X_1	X_2	Y
1	1	2	0
2	2	1	0
3	3	4	1
4	4	3	1
5	5	5	1
6	6	2	0

(a) Compute Gini impurity of the root.

(b) Evaluate all possible splits on X_1.

(c) Evaluate all possible splits on X_2.

(d) Choose the best split and compute impurity reduction.

(e) Grow the tree to depth 2.

(f) Hand-trace classification of point $(4, 2)$.

Problem 8: AdaBoost – Hand Tracing

Consider binary labels $y \in \{-1, 1\}$ with initial weights $w_i = 1/4$:

ID	X	Y
1	1	1
2	2	1
3	3	-1
4	4	-1

Weak learner: threshold classifier $h(x) = \text{sign}(x - t)$.

(a) Try all thresholds and compute weighted error.

(b) Compute α_1.

(c) Update weights explicitly.

(d) Normalize weights.

(e) Perform second boosting round.

(f) Write final strong classifier explicitly.

Problem 9: Gradient Boosting – Residual Tracing

Given regression dataset:

ID	X	Y
1	1	2
2	2	3
3	3	6
4	4	8

(a) Initialize f_0 as mean of Y.

(b) Compute residuals.

(c) Fit depth-1 regression tree manually.

(d) Update model with learning rate $\eta = 0.5$.

(e) Perform second boosting iteration.

(f) Compute final predictions.

Problem 10: Logistic Regression – Analytical and Numerical

Given dataset:

ID	x	y
1	0	0
2	1	0
3	2	1
4	3	1

Model:

$$p(y = 1|x) = \sigma(wx + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

(a) Write log-likelihood function.

(b) Derive gradient expressions.

(c) Perform one step of gradient ascent from $(w, b) = (0, 0)$.

(d) Compute Hessian matrix.

(e) Perform one Newton step.

(f) Hand-trace classification of $x = 1.5$.

Problem 11: Comparative Theoretical Analysis

(a) Compare impurity minimization vs exponential loss vs logistic loss.

(b) Show that AdaBoost minimizes exponential loss.

(c) Show that logistic regression minimizes log-loss.

(d) Show that gradient boosting performs functional gradient descent.

(e) Discuss bias-variance trade-offs analytically.

Supplementary Theory: Gini Score

The Gini impurity of a node with class probabilities $p_1, p_2, \ldots, p_K$ is defined as

$$G = 1 - \sum_{k=1}^{K} p_k^2.$$

For binary classification (p = proportion of class 1):

$$G = 2p(1 - p).$$

It measures the probability of misclassification when a label is drawn randomly according to class proportions.

(a) Prove that G is maximized when class proportions are equal.

(b) Show that $G = 0$ for a pure node.

(c) Compare Gini impurity with entropy analytically.

Problem 12: Weighted Logistic Regression

Consider weighted binary logistic regression with dataset:

ID	x	y	w
1	0	0	1
2	1	0	2
3	2	1	3
4	3	1	4

Model:

$$p(y=1|x)=\sigma(wx+b)$$

Weighted log-likelihood:

$$\ell(w,b)=\sum_{i=1}^{n} w_i\left[y_i\log\sigma(z_i)+(1-y_i)\log(1-\sigma(z_i))\right].$$

(a) Derive gradient expressions for w and b.

(b) Derive the Hessian matrix.

(c) Perform one weighted gradient ascent step from $(w,b)=(0,0)$.

(d) Explain how weighting alters decision boundary geometrically.

(e) Show relation to importance-weighted maximum likelihood.

Problem 13: Decision Tree for Regression (Numerical Tracing)

Consider regression dataset:

ID	X	Y
1	1	2
2	2	3
3	3	2
4	4	6
5	5	7
6	6	8

(a) Compute total variance (SSE) at root.

(b) Evaluate all possible splits and compute SSE for each.

(c) Select optimal first split.

(d) Grow tree to depth 2.

(e) Hand-trace prediction for $X = 4.5$.

(f) Compare variance reduction criterion with Gini impurity.

Problem 14: Logistic Regression via Newton's Method (Comprehensive Derivation and Numerical Tracing)

Consider the binary classification dataset with augmented feature vector $x_i = (1, x_{i1})^T$:

ID	x_{i1}	y_i
1	0	0
2	1	0
3	2	1
4	3	1
5	4	1

Model:

$$p_i = P(y_i = 1|x_i) = \sigma(z_i), \quad z_i = w^T x_i, \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

Let $X \in \mathbb{R}^{n \times 2}$ be the design matrix.

Part A: Likelihood and Convexity

(a) Write the log-likelihood function:

$$\ell(w) = \sum_{i=1}^{n} \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right].$$

(b) Show that $\ell(w)$ is concave by computing the Hessian.

(c) Prove that the optimization problem has a unique maximizer.

Part B: Gradient and Hessian Derivation

(a) Derive the gradient in matrix form:

$$\nabla \ell(w) = X^T (y - p).$$

(b) Show that the Hessian equals:

$$\nabla^2 \ell(w) = -X^T W X,$$

where $W = \mathrm{diag}(p_i(1 - p_i))$.

(c) Show explicitly that W is positive definite.

Part C: Newton Update

(a) Starting from Newton's general formula:

$$w_{k+1} = w_k - [\nabla^2 \ell(w_k)]^{-1} \nabla \ell(w_k),$$

show that

$$w_{k+1} = w_k + (X^T W X)^{-1} X^T (y - p).$$

(b) Rewrite the Newton step as the solution to a weighted least squares problem.

(c) Derive the Iteratively Reweighted Least Squares (IRLS) formulation explicitly by defining a working response variable.

Part D: Numerical Hand Tracing

Starting from initial parameter vector $w^{(0)} = (0, 0)^T$:

(a) Compute $p_i^{(0)}$ for all samples.

(b) Compute gradient vector numerically.

(c) Construct matrix $W^{(0)}$.

(d) Compute $X^T W^{(0)} X$ explicitly.

(e) Compute one full Newton update step.

(f) Compute updated probabilities after one iteration.

(g) Interpret geometrically how the decision boundary shifts.

Part F: Regularized Extension

Consider L2-regularized logistic regression:

$$\ell_{reg}(w) = \ell(w) - \frac{\lambda}{2} \|w\|^2.$$

(a) Derive regularized gradient.

(b) Derive regularized Hessian.

(c) Write Newton update for the regularized problem.

(d) Show how regularization improves conditioning.